Software Test Plan (STP)

Online Marketplace for Handcrafted Goods

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Roll Number	Student Name	Project Role	Testing & Verification Responsibility
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Document Revision History

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0.1	14-Oct-2026	Kishan K Shenoy	Initial test strategy outline and environment setup requirements	Draft Candidate

Version	Date	Author	Change Summary	Approval Status
0.9	25-Oct-2026	Team 08	Authored 30+ test cases covering positive, negative, and concurrency paths	Internal Review
1.0	27-Oct-2026	Team 08	Final release establishing full bidirectional RTM and 40 planned test cases	Submitted for Review (Pending Approval)

Document Review & Approval Sign-Off

Role	Name	Affiliation / Email	Sign-Off Status
Course Coordinator / Instructor	Chirag N. Bongale	PES University	Pending Review & Evaluation
Quality Assurance Lead	Kishan K Shenoy	PES1UG24AM138	Submitted by Author (27-Oct-2026)
Team Lead / Development Lead	Krrish Singla	PES1UG24AM141	Submitted by Author (27-Oct-2026)

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1. Introduction

1.1 Purpose & Test Objectives

This Software Test Plan (STP) defines the verification and validation strategy for the "Online Marketplace for Handcrafted Goods", developed by Team 08 for the Software Engineering Mini Project at PES University. It details test items, approach levels, environment configurations, milestone schedules, and an exhaustive suite of planned test cases covering functional, security, concurrency, performance, accessibility, and CI/CD criteria. Note: This document represents a formal testing specification for planned execution across Sprint 1 (27–31 Oct 2026) and Sprint 2 (2–7 Nov 2026), and does not claim completed execution prior to software implementation.

1.2 Scope & Boundaries

Testing scope encompasses all software tiers: Client Web UI, RESTful API endpoints, security middleware, domain services, relational persistence layer, and external integrations (payment gateway stub, SMTP relay, and ReportLab PDF engine). External banking core networks and physical package shipment providers are excluded.

1.3 Companion Specifications & References

- **SRS_Team08_OnlineMarketplace.docx / .pdf**: Requirements baseline defining all 20 FRs, 5 NFRs, and 5 SRs.
- **SAD_Team08_OnlineMarketplace.docx / .pdf**: Architecture specification defining component interfaces, sequence diagrams, and ERD.
- **GitHub Repository**: https://github.com/krrishsingla11/SE-LAB-TEAM-08 (planned test code repository).

1.4 Definitions and Acronyms

Acronym / Term	Definition in Testing Context
STP	Software Test Plan: this verification and validation plan.
TC	Test Case: unique identifier for an individual planned verification procedure.
RTM	Requirements Traceability Matrix: mapping test cases to SRS requirements and SAD components.
UAT	User Acceptance Testing: end-to-end scenario validation against business personas.
VU	Virtual Users: simulated concurrent HTTP threads generated during load testing.
CI/CD	Continuous Integration / Continuous Deployment: automated test runner in GitHub Actions.

2. Test Items & Modules Under Test

Software Test Item / Module	Subsystems and Capabilities Under Verification
Identity & Authentication Module	User registration, bcrypt password hashing, JWT creation/validation, RBAC middleware, password reset.
Catalog & Listing Module	Product creation, multi-image upload, artisan ownership verification, catalog search, category filtering.
Shopping Cart & Checkout Module	Cart persistence, dynamic quantity updates, pre-checkout inventory validation, shipping address input.
Payment Processing Module	Payment initiation, mock stub communication (Sprint 1), card decline handling, idempotent retry logic.
Order Fulfillment Module	Transactional order creation, status progression (Pending->Processing->Shipped->Delivered), customer email triggers.
Inventory Locking Module	Atomic stock decrement via row-level locks (SELECT ... FOR UPDATE), low-stock alert notifications.
Invoice Generation Module	Server-side PDF compilation via ReportLab, static file caching, authenticated customer download.
Reporting & Analytics Module	Artisan monthly revenue aggregation, units sold computation, CSV and PDF export streaming.
Cross-Cutting Security & Performance	OWASP ZAP vulnerability checks, rate limiting (5 attempts/10 min), 100 VU load test, 500 VU stress test.
Automated CI/CD Pipeline	GitHub Actions workflow executing flake8 linting, pytest suites, and Docker health-checks.

3. Features to be Tested

All 20 functional requirements (OMH-F-001 to OMH-F-020), 5 non-functional requirements (OMH-NF-001 to OMH-NF-005), and 5 security requirements (OMH-SR-001 to OMH-SR-005) are scheduled for rigorous verification across the test suites.

4. Features Excluded from Testing

- **External Payment Gateway Settlement**: Inter-bank fund transfers and card brand clearance networks are assumed validated by the gateway provider.
- **Physical Shipping Logistics**: Courier vehicle routing, parcel handoff, and physical barcode scanning are external to software boundaries.
- **Third-Party Email Delivery Infrastructure**: Actual delivery to external recipient mail servers is stubbed using Mailtrap in development.

5. Test Approach & Verification Strategy

5.1 Test Levels

- **Unit Testing (pytest)**: Validates isolated functions, business logic invariants, and validation rules with mocked database sessions.
- **Integration Testing**: Validates interaction between service classes, SQLAlchemy ORM, and a containerized test database instance.
- **System Testing (Postman / REST)**: Executes complete end-to-end HTTP API workflows against a fully deployed multi-container Docker stack.
- **User Acceptance Testing (UAT)**: Validates user journeys (Buyer purchasing craft, Artisan publishing listing and exporting report) in real browsers.

5.2 Test Types & Quality Dimensions

- **Functional Testing**: Verifies that every requirement produces exact expected outputs for valid (positive), invalid (negative), and edge (boundary) inputs.
- **Concurrency & Race Condition Testing**: Executes multi-threaded checkout scripts simulating simultaneous purchases of the final item.
- **Performance & Load Testing**: Utilizes Apache JMeter to generate 100 concurrent virtual users measuring 95th percentile API response times.
- **Security Testing**: Automated OWASP ZAP vulnerability scanning for SQLi/XSS, manual JWT tampering, and rate limiter verification.
- **Accessibility Testing**: Automated axe-core browser audit verifying WCAG 2.1 AA compliance and screen-reader accessibility.

5.3 Security Verification Strategy

Security testing rigorously verifies: (1) TLS 1.2+ mandatory enforcement (TC-SEC-01); (2) Zero raw cardholder data (PAN/CVV) storage or logging (TC-SEC-02); (3) bcrypt work factor >= 12 password hashing without plaintext storage (TC-SEC-03); (4) JWT HMAC-SHA256 signature and 24h expiration enforcement (TC-SEC-04); (5) ORM parameterized input sanitization against SQL injection and XSS (TC-SEC-05); (6) Authentication rate limiting blocking IP after 5 failed logins (TC-SEC-06); and (7) Immutable security audit logging with sensitive data masking (TC-SEC-07).

5.4 Entry and Exit Criteria

- **Test Execution Entry Criteria**: 1. Stable build containerized via Docker Compose; 2. Test database seeded with baseline fixtures; 3. Unit test suite passes with 0 failures; 4. Mock payment stub operational.
- **Sprint Sign-Off Exit Criteria**: 1. 100% of planned test cases executed; 2. Zero open Critical or High severity defects; 3. 95th percentile response time <= 500 ms; 4. OWASP ZAP reports 0 high-risk vulnerabilities.

6. Test Environment & Harness Architecture

Environment Component	Specification & Tooling Configuration
Developer Workstations	Windows 11 / Linux (Ubuntu 22.04 LTS), 16GB RAM, Docker Desktop 26.x.
CI/CD Runner Host	GitHub Actions hosted runner (ubuntu-latest), Docker Compose v2.27.
Application Container	Python 3.12-slim base image, Flask/FastAPI REST application on port 5000.
Database Container	MySQL 8.0 / PostgreSQL 15 container initialized with Alembic schema migrations.
Mock Payment Service	Local Python HTTP stub responding with preconfigured 200 OK or 402 Declined payloads.
Email Sandbox	Mailtrap SMTP sandbox capturing notification emails without external transmission.
Automated Test Tools	pytest 8.x, pytest-cov, Postman 10.x, Apache JMeter 5.6, OWASP ZAP 2.14, axe-core.

7. Test Schedule & Course Milestones

All test milestones strictly follow the instructor announcement (authoritative schedule):

Course Milestone / Test Phase	Scheduled Dates	Responsible	Planned Deliverable & Verification Scope
Deliverables Submission (SRS, SAD, STP)	27-Oct-2026	Team 08	Submission of all three documentation deliverables before Sprint 1.
Sprint 1 Development & Testing Start	27-Oct-2026	All Members	Execute unit tests and API tests for Auth, Catalog, Cart, and Payment stub.
Sprint 1 Test Execution Freeze	31-Oct-2026	Kishan K Shenoy	Verify TC-AUTH-01..TC-AUTH-07, TC-PROD-01..TC-PROD-05, TC-CART-01..TC-PAY-03.
Sprint 1 Outcome Video Upload	31-Oct-2026	All Members	1-minute video demonstrating working product committed to GitHub repo.
Sprint 2 Development & Testing Start	02-Nov-2026	All Members	Execute tests for Orders, Inventory Locks, Invoices, Reports, and Security.
Sprint 2 Full Test & Load Freeze	07-Nov-2026	Kishan K Shenoy	Full regression suite, JMeter load tests, OWASP ZAP scan, UAT sign-off.
Sprint 2 Outcome Video Upload	07-Nov-2026	All Members	2-minute video demonstrating complete working product uploaded to repo.
Documentation & Code Freeze	07-Nov-2026	Team 08	Freeze development cycle; document any project deviations.

8. Test Deliverables & Artifacts

- **Software Test Plan (STP v1.0)**: This formal planning deliverable.
- **Automated Test Scripts**: pytest unit/integration test suite located in /tests directory.
- **Postman API Test Collection**: JSON collection containing 30+ automated request assertions.
- **JMeter Performance Test Plan**: marketplace_load_test.jmx simulating 100 and 500 concurrent users.
- **OWASP ZAP Vulnerability Report**: Automated security scan summary report.
- **Final Test Summary Report**: Summary of test execution results, pass/fail counts, and defect log.

9. Roles and Responsibilities

- **Kishan K Shenoy (QA Lead / Test Engineer)**: Author of STP; coordinates test execution, runs JMeter and OWASP ZAP scans, manages defect logging in Jira.
- **Krrish Singla (Team Lead / Dev Lead)**: Supervises CI/CD pipeline integration, enforces PR code reviews, triages critical defects.
- **Kushal G (Backend & Database Engineer)**: Configures Docker test harness, seeds test database fixtures, resolves database locking defects.
- **Namith Kumar C (Frontend & UI/UX Engineer)**: Validates client forms, executes cross-browser tests, runs axe-core accessibility audits.
- **Chirag N. Bongale (Course Coordinator)**: Reviews deliverables and evaluates sprint demonstration videos submitted to the project repository.

10. Risks and Mitigation Strategies

- **Risk 1: Flaky Integration Tests Due to Database State**: Mitigation: Each pytest run executes inside an isolated Docker container with automated database rollback.
- **Risk 2: CI/CD Pipeline Execution Timeouts**: Mitigation: Cache pip dependencies in GitHub Actions; decouple long-running load tests from the standard pull request gate.
- **Risk 3: Concurrency Race Conditions Missed in Manual Testing**: Mitigation: Dedicated multi-threaded Python stress script (TC-INV-01) simulating concurrent checkouts.

11. Assumptions & Environmental Dependencies

Assumes Docker Desktop is available on all developer machines and GitHub Actions minutes are active.

12. Suspension & Resumption Criteria

- **Suspension Criteria**: Testing is suspended if the Docker environment fails to boot or if > 30% of baseline unit tests fail.
- **Resumption Criteria**: Testing resumes once blocking defects are resolved and verified by the Development Lead.

13. Detailed Test Cases (40 Planned Test Scenarios)

This section details 40 comprehensive, planned test cases spanning all functional and non-functional requirements. Each test case specifies a unique ID, SRS requirement mapping, priority, execution preconditions, exact test steps, and expected verification results. All statuses are PLANNED.

13.1 Authentication & RBAC Test Cases

TC ID	Scenario & Priority	SRS Ref	Preconditions & Test Data	Test Procedure Steps	Expected Verification Result	Status
TC-AUTH-01	Positive: Valid Buyer Registration (High)	OMH-F-001	DB running; email not in DB. Data: test_buyer@pes.edu, pass: StrongP@ss123	1. POST /api/auth/register with valid email, name, role=buyer. 2. Inspect response status and DB record.	HTTP 201 Created; user row in DB; password hashed with bcrypt; welcome email in Mailtrap.	Planned
TC-AUTH-02	Negative: Duplicate Email Registration (High)	OMH-F-001	User test_buyer@pes.edu already exists in DB.	1. POST /api/auth/register with identical email test_buyer@pes.edu.	HTTP 409 Conflict; JSON error "Email already registered"; no duplicate row created.	Planned
TC-AUTH-03	Positive: Successful Login & JWT Token (High)	OMH-F-002	Registered active user test_buyer@pes.edu in DB.	1. POST /api/auth/login with valid credentials. 2. Decode returned JWT payload.	HTTP 200 OK; JWT token returned; payload contains user_id, role="buyer", exp=24 hours.	Planned
TC-AUTH-04	Negative: Login with Incorrect Password (High)	OMH-F-002	User test_buyer@pes.edu exists.	1. POST /api/auth/login with wrong password "WrongPass999".	HTTP 401 Unauthorized; JSON error "Invalid credentials"; zero token issued.	Planned
TC-AUTH-05	Security: RBAC Denial on Protected Route (High)	OMH-F-003	Authenticated Buyer with valid JWT (role="buyer").	1. GET /api/seller/dashboard sending Buyer JWT in Authorization header.	HTTP 403 Forbidden; error "Artisan privilege required"; access strictly blocked.	Planned
TC-AUTH-06	Positive: Password Reset Flow (Medium)	OMH-F-004	Registered user account test_buyer@pes.edu.	1. POST /api/auth/reset-password. 2. Extract token from Mailtrap. 3. POST /api/auth/reset-password/confirm with token & new pass.	HTTP 200 OK; password updated in DB; subsequent login with new password succeeds.	Planned
TC-AUTH-07	Boundary: Password Reset Expired Token (Medium)	OMH-F-004	Reset token created > 15 minutes ago.	1. Submit expired token to /api/auth/reset-password/confirm.	HTTP 400 Bad Request; error "Reset token has expired"; password unchanged.	Planned

13.2 Product Catalog & Search Test Cases

TC ID	Scenario & Priority	SRS Ref	Preconditions & Test Data	Test Procedure Steps	Expected Verification Result	Status
TC-PROD-01	Positive: Artisan Creates Handcrafted Listing (High)	OMH-F-005	Authenticated Artisan JWT; image files available.	1. POST /api/products multipart with title="Silk Scarf", price=450.00, stock=20, 2 image JPGs.	HTTP 201 Created; product_id returned; images saved to product_images table; inventory set to 20.	Planned
TC-PROD-02	Negative/Boundary: Listing with Negative Price (High)	OMH-F-005	Authenticated Artisan JWT.	1. POST /api/products with price=-50.00 and stock_qty=-5.	HTTP 422 Unprocessable Entity; validation error identifying invalid price and stock fields.	Planned
TC-PROD-03	Positive: Artisan Updates Own Listing (High)	OMH-F-006	Product #101 owned by authenticated Artisan.	1. PUT /api/products/101 with {"price": 499.00, "stock_qty": 35}.	HTTP 200 OK; product price updated to 499.00; inventory row updated to 35 in DB.	Planned
TC-PROD-04	Security: Modify Another Artisan's Product (High)	OMH-F-006	Artisan A logged in; Product #102 owned by Artisan B.	1. PUT /api/products/102 with price update payload.	HTTP 403 Forbidden; error "Not authorized to modify this listing"; data untouched.	Planned

TC ID	Scenario & Priority	SRS Ref	Preconditions & Test Data	Test Procedure Steps	Expected Verification Result	Status
TC-PROD-05	Positive: Catalog Keyword Search & Filter (High)	OMH-F-007	10 active crafts in DB across Pottery and Textiles.	1. GET /api/products?q=ceramic&category=1. 2. Inspect returned array.	HTTP 200 OK; returns only active pottery products containing keyword "ceramic".	Planned

13.3 Shopping Cart & Pre-Checkout Test Cases

TC ID	Scenario & Priority	SRS Ref	Preconditions & Test Data	Test Procedure Steps	Expected Verification Result	Status
TC-CART-01	Positive: Add Item & Update Cart Quantity (High)	OMH-F-008	Buyer logged in; Product #101 has stock=20.	1. POST /api/cart/items with product_id=101, qty=2. 2. PUT /api/cart/items/101 with qty=3.	HTTP 200 OK; cart subtotal dynamically recalculates to 3 * unit_price; persisted in DB.	Planned
TC-CART-02	Negative: Add Quantity Exceeding Stock (High)	OMH-F-008	Product #101 stock=5.	1. POST /api/cart/items with product_id=101, qty=10.	HTTP 400 Bad Request; error "Requested quantity exceeds available inventory (5)".	Planned
TC-CART-03	Boundary: Remove Last Item from Cart (Medium)	OMH-F-008	Cart with exactly 1 line item.	1. DELETE /api/cart/items/101. 2. GET /api/cart.	HTTP 200 OK; cart item deleted; cart items array empty; subtotal equals 0.00.	Planned
TC-CHK-01	Negative/Boundary: Pre-Checkout Stock Depleted (High)	OMH-F-009	Cart contains Product #103 qty=2; stock reduced to 0.	1. POST /api/checkout with delivery address.	HTTP 400 Bad Request; checkout aborted BEFORE payment; error identifies Product #103 as out of stock.	Planned

13.4 Payment Processing & Retry Test Cases

TC ID	Scenario & Priority	SRS Ref	Preconditions & Test Data	Test Procedure Steps	Expected Verification Result	Status
TC-PAY-01	Positive: Payment Success via Mock Stub (High)	OMH-F-010	Stock reserved; valid mock payment token "tok_visa_success".	1. POST /api/checkout with tok_visa_success.	HTTP 201 Created; payment captured; order fulfillment status set to "Processing"; txn_ref "TXN-MOCK-OK" saved.	Planned
TC-PAY-02	Negative: Payment Declined by Gateway (High)	OMH-F-010	Stock reserved; declined token "tok_charge_declined".	1. POST /api/checkout with tok_charge_declined.	HTTP 402 Payment Required; error "Card declined"; DB rolls back stock lock; zero order created.	Planned
TC-PAY-03	Boundary/Retry: Idempotent Payment Retry (Medium)	OMH-F-010	Payment timed out; client submits identical Idempotency-Key.	1. Re-POST /api/checkout with same idempotency key.	HTTP 200/201 returns original transaction outcome; zero duplicate charges or orders created.	Planned

13.5 Order Management & Tracking Test Cases

TC ID	Scenario & Priority	SRS Ref	Preconditions & Test Data	Test Procedure Steps	Expected Verification Result	Status
TC-ORD-01	Positive: Transactional Order Creation (High)	OMH-F-012	Payment approved; valid cart items.	1. Complete checkout; inspect DB orders, order_items tables.	Order record created with total, shipping address, status="Processing", payment_status="captured"; cart items cleared.	Planned

TC ID	Scenario & Priority	SRS Ref	Preconditions & Test Data	Test Procedure Steps	Expected Verification Result	Status
TC-ORD-02	Positive: Artisan Progresses Order Status (High)	OMH-F-013	Order #55 status="Processing" owned by Artisan.	1. PUT /api/orders/55/status with status="Shipped". 2. PUT with status="Delivered".	HTTP 200 OK; status transitions correctly; buyer views updated tracking in order history.	Planned
TC-ORD-03	Positive: Order Event Email Dispatched (Medium)	OMH-F-014	Order status updated to "Shipped".	1. Trigger status update. 2. Inspect Mailtrap inbox for buyer email.	Templated email received containing order ID, tracking status, and delivery address.	Planned

13.6 Inventory Locking & Stock Alerts Test Cases

TC ID	Scenario & Priority	SRS Ref	Preconditions & Test Data	Test Procedure Steps	Expected Verification Result	Status
TC-INV-01	Concurrency: Atomic Stock Decrement Lock (High)	OMH-F-015	Product #105 has stock_quantity = 1. Two buyers A and B in cart.	1. Simultaneously execute 2 threads calling POST /api/checkout for Product #105.	Thread 1 gets HTTP 201 (Order Processing, Stock Decremented); Thread 2 gets HTTP 400 (Out of stock); stock=0, never negative.	Planned
TC-INV-02	Positive: Low-Stock Dashboard Alert (Medium)	OMH-F-016	Product threshold=5; stock decrements from 6 to 5.	1. Complete order buying 1 unit. 2. GET /api/seller/dashboard.	Dashboard notifications include low-stock alert badge for product with remaining count 5.	Planned

13.7 Server-Side PDF Invoicing Test Cases

TC ID	Scenario & Priority	SRS Ref	Preconditions & Test Data	Test Procedure Steps	Expected Verification Result	Status
TC-INV-03	Positive: PDF Invoice Generation & Formatting (High)	OMH-F-017	Completed order #55 in DB.	1. Inspect server storage for order #55 invoice PDF. 2. Parse PDF with PyMuPDF.	PDF file exists; contains Order ID, Buyer Name, Itemized Line Items, Tax, and Total Amount.	Planned
TC-INV-04	Security/Negative: Non-Owner Download Blocked (Medium)	OMH-F-018	Order #55 belongs to Buyer A; Buyer B logged in.	1. GET /api/orders/55/invoice with Buyer B JWT token.	HTTP 403 Forbidden; error "Not authorized to access this invoice"; PDF download blocked.	Planned

13.8 Sales Analytics & Report Export Test Cases

TC ID	Scenario & Priority	SRS Ref	Preconditions & Test Data	Test Procedure Steps	Expected Verification Result	Status
TC-RPT-01	Positive: Monthly Sales Analytics Aggregation (High)	OMH-F-019	3 completed orders for Artisan in October 2026 ($1200 total).	1. GET /api/reports/sales?period=monthly.	HTTP 200 OK; JSON payload returns revenue=1200.00, order_count=3, correct item breakdown.	Planned
TC-RPT-02	Positive: Export Sales Report to CSV & PDF (Medium)	OMH-F-020	Sales data present for artisan.	1. GET /api/reports/export?format=csv. 2. GET /api/reports/export?format=pdf.	HTTP 200 OK with Content-Type: text/csv and application/pdf; files stream without corruption.	Planned

13.9 Comprehensive Security Test Cases (TC-SEC-01 to TC-SEC-07)

TC ID	Scenario & Priority	SRS Ref	Preconditions & Test Data	Test Procedure Steps	Expected Verification Result	Status

TC ID	Scenario & Priority	SRS Ref	Preconditions & Test Data	Test Procedure Steps	Expected Verification Result	Status
TC-SEC-01	Security: TLS 1.2+ Mandatory & HTTP 301 Redirect (High)	OMH-NF-003	Marketplace web server running with TLS certificate.	1. Send plaintext HTTP GET request to port 80. 2. Send HTTPS request with TLS 1.0/1.1. 3. Send HTTPS with TLS 1.2+.	Plaintext request returns 301 redirect to https://; TLS 1.0/1.1 handshake rejected; TLS 1.2/1.3 accepted.	Planned
TC-SEC-02	Security/PCI: Zero Raw Cardholder Data Persisted (High)	OMH-F-011	Complete successful checkout flow using payment stub.	1. Execute SQL dump of all database tables (users, orders, payments, logs). 2. Grep for 16-digit card patterns and CVV.	Zero credit card PAN numbers or CVV codes found in DB or log files; only opaque payment tokens stored.	Planned
TC-SEC-03	Security: bcrypt Password Hash & Zero Plaintext (High)	OMH-SR-001	New user registered via POST /api/auth/register.	1. Query users table for registered account. 2. Verify password_hash column value format and length.	Password stored strictly as bcrypt hash starting with $2b$12$; zero plaintext passwords in DB or logs.	Planned
TC-SEC-04	Security: JWT Signature & 24h Expiry Enforcement (High)	OMH-SR-002	User logged in; valid JWT token obtained.	1. Tamper with token payload (change role to admin). 2. Present expired token (>24h). 3. Send to protected endpoint.	Tampered token fails HMAC signature verification (401); expired token returns HTTP 401 Unauthorized.	Planned
TC-SEC-05	Security: Input Sanitisation (SQLi & XSS Defense) (High)	OMH-SR-003	Marketplace API deployed in test environment; OWASP ZAP configured.	1. Inject SQLi payloads ("' OR 1=1--") into search query params. 2. Submit script tags into product description. 3. Run ZAP scan.	SQL payloads treated as literal text; script tags escaped by UI template; ZAP reports 0 high/critical vulnerabilities.	Planned
TC-SEC-06	Security: Rate Limiting Blocks Brute Force (429) (High)	OMH-SR-004	Rate limiter active on /api/auth/login.	1. Send 6 consecutive failed login requests from identical IP within a 2-minute window.	Requests 1 through 5 return HTTP 401; 6th request is blocked and returns HTTP 429 Too Many Requests.	Planned
TC-SEC-07	Security: Structured Audit Logging & Masking (High)	OMH-SR-005	Execute login, product creation, and order checkout.	1. Inspect application log output on stdout. 2. Verify JSON schema, UTC timestamps, user IDs, and data masking.	Logs contain structured JSON with UTC timestamps and user IDs; sensitive fields (passwords, tokens) are masked.	Planned

13.10 Performance, Scalability, Accessibility & Reliability Tests

TC ID	Scenario & Priority	SRS Ref	Preconditions & Test Data	Test Procedure Steps	Expected Verification Result	Status
TC-PERF-01	Performance: API Latency Under Nominal Load (100 VU) (High)	OMH-NF-001	Multi-container stack; JMeter installed.	1. Execute 100 VU load test across search and cart endpoints for 5 minutes.	95th percentile response latency <= 500 ms; HTTP error rate = 0.00%.	Planned
TC-PERF-02	Scalability: 500 Concurrent Session Stress Test (Medium)	OMH-NF-004	Multi-container stack.	1. Execute 500 VU stress test with ramp-up over 2 minutes.	System sustains load without server crashes or 502/503 errors; DB connection pool remains stable.	Planned
TC-UX-01	Usability: WCAG 2.1 AA Accessibility Audit (Medium)	OMH-NF-005	Web frontend running locally.	1. Execute axe-core accessibility scanner on landing, product, and checkout pages.	Zero critical or serious accessibility violations; color contrast >= 4.5:1 confirmed.	Planned
TC-OPS-01	Reliability: Health Monitor & 99.5% Service Uptime (High)	OMH-NF-002	Deployed application in container environment.	1. Configure automated health-check ping against /api/health every 60 seconds over operational test window.	HTTP 200 OK returned on >= 99.5% of pings; database connectivity verified healthy.	Planned
TC-CICD-01	CI/CD: Automated Build, Lint & Pytest on Pull Request (High)	Planned CI/CD	GitHub repo with .github/workflows/ci.yml.	1. Push branch and open Pull Request on GitHub.	GitHub Actions workflow triggers automatically; executes flake8 linting, runs pytest suite, and passes.	Planned

14. Requirements Traceability Matrix (RTM)

The RTM maps all requirements from the SRS and SAD to their corresponding STP test cases. In accordance with course guidelines, all verification statuses are designated as PLANNED prior to active sprint test execution.

Req ID	Requirement Description	Test Case IDs	Test Level	Sprint	Status
OMH-F-001	User Registration & Email Uniqueness	TC-AUTH-01, TC-AUTH-02	Unit / Integration	1	Planned
OMH-F-002	Login & 24h JWT Issuance	TC-AUTH-03, TC-AUTH-04	Unit / System	1	Planned
OMH-F-003	RBAC Route & View Enforcement	TC-AUTH-05	System / Security	1	Planned
OMH-F-004	Password Reset Token Flow	TC-AUTH-06, TC-AUTH-07	Integration / System	1	Planned
OMH-F-005	Artisan Product Listing Creation	TC-PROD-01, TC-PROD-02	Integration / System	1	Planned
OMH-F-006	Artisan Listing Modification/Deletion	TC-PROD-03, TC-PROD-04	Integration / System	1	Planned
OMH-F-007	Catalog Browsing, Filter & Search	TC-PROD-05	System / UAT	1	Planned
OMH-F-008	Persistent Shopping Cart Operations	TC-CART-01, TC-CART-03	Integration / System	1	Planned
OMH-F-009	Pre-Checkout Stock Validation	TC-CART-02, TC-CHK-01	Integration / System	1	Planned
OMH-F-010	Payment Processing (Mock/Gateway)	TC-PAY-01, TC-PAY-02, TC-PAY-03	Integration / System	1/2	Planned
OMH-F-011	Prohibit Raw Cardholder Data Storage	TC-SEC-02	Security Audit	1/2	Planned
OMH-F-012	Transactional Order Creation	TC-ORD-01	Integration / System	2	Planned
OMH-F-013	Artisan Order Status Transitions	TC-ORD-02	System / UAT	2	Planned
OMH-F-014	Order Notification Email Triggers	TC-ORD-03	Integration / System	2	Planned
OMH-F-015	Concurrency-Safe Atomic Stock Lock	TC-INV-01	Concurrency / Stress	2	Planned
OMH-F-016	Low-Stock Dashboard Alert System	TC-INV-02	Integration / System	2	Planned
OMH-F-017	Server PDF Invoice Compilation	TC-INV-03	Integration / System	2	Planned
OMH-F-018	Customer PDF Invoice Retrieval	TC-INV-04	System / Security	2	Planned
OMH-F-019	Artisan Sales Revenue Analytics	TC-RPT-01	Integration / System	2	Planned
OMH-F-020	Export Sales Reports (CSV/PDF)	TC-RPT-02	System / UAT	2	Planned
OMH-NF-001	Response Time <= 500 ms (100 VU)	TC-PERF-01	Performance Load	2	Planned
OMH-NF-002	99.5% Service Availability	TC-OPS-01	Infrastructure Monitor	2	Planned
OMH-NF-003	Mandatory TLS 1.2+ HTTPS	TC-SEC-01	Security / Infra	1	Planned
OMH-NF-004	500 Concurrent Session Scalability	TC-PERF-02	Stress Testing	2	Planned
OMH-NF-005	WCAG 2.1 AA Usability Compliance	TC-UX-01	Accessibility / UAT	2	Planned
OMH-SR-001	bcrypt Password Hash Verification	TC-SEC-03	Security Unit	1	Planned
OMH-SR-002	JWT 24-Hour Expiration Check	TC-SEC-04	Security System	1	Planned
OMH-SR-003	Input Sanitization (SQLi / XSS)	TC-SEC-05	Security Scan	2	Planned
OMH-SR-004	Login Rate Limiter (5 attempts/10m)	TC-SEC-06	Security System	2	Planned
OMH-SR-005	Structured Immutable Audit Logging	TC-SEC-07	Integration Audit	2	Planned

15. Test Metrics, Defect Handling & Reporting

- **Test Execution Metrics**: Total Planned vs Executed Test Cases; Pass/Fail Percentage; Requirement Coverage Ratio.
- **Defect Severity Classification**: Critical (Blocks checkout or corrupts data); High (Major feature failure); Medium (Non-blocking edge case); Low (Cosmetic/UI).
- **Defect Lifecycle in Jira**: New -> In Progress -> Ready for Retest -> Closed / Verified.

16. Formal Approvals & Sign-Offs

Project Role	Sign-Off Name	Institutional Email / ID	Date & Sign-Off Status
Course Coordinator / Instructor	Chirag N. Bongale	PES University	Pending Evaluation & Review
Quality Assurance Lead	Kishan K Shenoy	PES1UG24AM138	Submitted by Author (27-Oct-2026)
Team Lead / Development Lead	Krrish Singla	PES1UG24AM141	Submitted by Author (27-Oct-2026)